



EDGE - AI INDUSTRIAL SAFETY



VISIONSAFE 360

Turning the cameras you **already own** into an **active line of defense** — stopping the accident **before** it happens.

The first unified safety ecosystem that fuses PPE compliance, forklift-worker proximity, fall detection, and ergonomic risk into one real-time platform — all running on-site at the edge.

PPE Compliance

Forklift Proximity

Fall Detection

Ergonomics · RULA/REBA

Mobile Alerts

4-in-1

Hazard models unified in a single platform

0 cloud

Video never leaves the facility — full edge privacy

~3s

From hazard to supervisor's phone

2 langs

Bilingual EN / AR · WCAG accessible

01 The Problem & Our Goal

WHY IT MATTERS

In industry, **the cost of being late is measured in lives**. Most workplaces still rely on cameras that only **record** what already happened. VisionSafe 360 makes that same infrastructure **recognize danger as it unfolds** and intervene before the accident occurs — replacing fragmented, reactive tools with continuous, evidence-driven prevention.

02 Value & Feasibility

BUILT FOR ADOPTION

Nothing to replace. It runs on the CCTV a site already operates. All analysis happens **locally at the edge** — no cloud-upload costs, data sovereignty satisfied, real-time response independent of the network. Scales linearly per camera, from one device to a multi-site enterprise, aligned with **OSHA, NIOSH & ISO 45001**.

03 How It Works

FOUR COHESIVE LAYERS



Unified AI Safety Intelligence

PPE, proximity, falls & posture monitored together — the central technical contribution.



Real-Time Command Center

Multi-camera dashboard, alert triage, safety zones & full incident lifecycle + mobile push.



Intelligent Decision Support

Detections validated before alerting — eliminating alert fatigue; analytics surface hazard patterns.



Resilient & Accessible

Keeps detecting offline, auto-syncs on reconnect; bilingual EN/AR, fully accessible.

04 Who Benefits

WHEREVER WORKERS & MACHINES SHARE SPACE

Manufacturing & Assembly

Warehouses & Logistics

Construction Sites

Oil, Gas & Heavy Industry

Ports · Mining · Quarrying

HSE & Safety Officers

Floor Supervisors

Workers

Regulators & Insurers

The **strongest beneficiaries are the workers themselves** — faster emergency response, protection from the most common hazards, and ergonomic intervention before chronic injury.

05 Expected Impact

FROM REACTIVE TO PREVENTIVE



Operational

Response time collapses from minutes/hours to **seconds** — the difference between recovery and tragedy.



Financial

Turns hidden injury costs — downtime, penalties, premiums, litigation — into preventable expense.



Organizational & Cultural

Every alert logged with evidence & audit trail — shifting from blame-based to **evidence-based safety**.

STUDENT TEAM

01 Hisham Mohamed

02 Mohamed Soltan

03 Raneem Mohamed

04 John Amin

05 Shams Elden Mohammed

SUPERVISOR

Supervisor Dr. Amr Ibrahim

Supervisor Eng. Aya Adel